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Graph Canonical Coherence Analysis

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Abstract

We propose graph canonical coherence analysis (gCChA), a novel framework that extends canonical correlation analysis to multivariate graph signals in the graph frequency domain. The proposed method addresses challenges posed by the inherent features of graphs: discreteness, finiteness, and irregularity. It identifies pairs of canonical graph signals that maximize their coherence, enabling the exploration of relationships between two sets of graph signals from a spectral perspective. This framework shows how these relationships change across different structural scales of the graph. We demonstrate the usefulness of this method through a simulation study and an application to economic and energy datasets of G20 countries.

Keywords: Canonical correlation analysis; Dimension reduction; Frequency domain; Graph signal processing; Multivariate graph signal.

1 Introduction

Canonical correlation analysis (CCA), originally proposed by Hotelling (1935, 1936), is a classical multivariate method for quantifying a linear relationship between two sets of multidimensional variables. Classical CCA deals with multidimensional data in Euclidean space, where each set of variables is represented as a vector. Brillinger (2001) further extended CCA to analyze two multivariate time series by formulating it in the frequency domain. However, in many fields, data are observed on irregular domains with graph structures, known as graph signals (Leus et al. 2023, Shuman et al. 2013), highlighting the need to develop CCA tailored to such domains. Specifically, univariate graph signals are scalar values located on the nodes of a graph, while multivariate graph signals consist of vector-valued data associated with each node, as shown in Figure 1 1. In the figure, the height and direction (upward or downward) of each bar indicate the magnitude and sign of the signal value mapped to each node, and bars with the same color correspond to the same attribute in the multivariate case. Graph signals defined on a path or cycle graph can be viewed as special cases that correspond to time series.

Developing CCA for multivariate graph signals presents unique challenges due to the inherent discreteness, finiteness, and irregularity of graphs. In this study, we propose a new CCA framework for graph signals called *graph canonical coherence analysis (gCChA)*. This framework aims to discover relationships between two multivariate graph signals on a common underlying graph. To account for the distinctive characteristics of the graph domain, we develop a framework in the graph frequency domain that bridges time-domain and graph-domain analyses, drawing on the frequency-domain approach of Brillinger (2001). Unlike several previous works that operate in the node domain, gCChA provides a systematic way to describe the relationships between two multivariate graph signals using low-dimensional canonical graph signals and to analyze these relationships from a spectral perspective across graph frequencies.

The R codes and datasets used in the numerical experiments are available at <https://github.com/qsoon/gCChA>.

2 Background

2.1 Graph Signals and Graph Frequency Domain

Let $\mathcal{G} = (\mathcal{V}, E, W)$ denote a weighted graph, possibly directed, where $\mathcal{V}$ is the set of n nodes, E is the set of edges, and W is the weighted adjacency matrix. We assume that $\mathcal{G}$ is finite, connected, and simple, meaning it has no self-loops or multiple edges. The nonnegative element w_{ij} of W represents the weight of the edge between nodes i and j , and a value of $w_{ij} = 0$ indicates no direct connection between them. A graph signal assigns a scalar or vector to each node of $\mathcal{G}$. A univariate graph signal can be viewed as a function $x: \mathcal{V} \rightarrow \mathbb{R}$ (or $\mathbb{C}$), while a multivariate graph signal takes values in higher-dimensional codomains. In the univariate case, the signal is represented by a vector x of size n , whose i th entry corresponds to the value at node i .

A key concept in graph signal processing (GSP) is the graph shift operator (GSO), denoted by S , which defines local interactions between neighboring nodes. Multiplying S by a univariate signal $x \in \mathbb{R}^n$ (or $\mathbb{C}^n$) produces a new signal whose value at each node is a weighted sum of the signal values at its neighbors (Sandryhaila and Moura 2013, Segarra et al. 2018). Common choices for GSOs include the weighted adjacency matrix and the graph Laplacian. For an undirected graph, the Laplacian is defined as $L = \text{diag}(\{\sum_{j=1}^n w_{ij}\}_{i=1}^n) - W$, where w_{ij} is the (i, j) th element of W . In

general settings, the GSO S is assumed to be normal, allowing it to be decomposed as $S = V\Lambda V^H$, where $V = (v_1 \mid \dots \mid v_n)$ contains orthonormal eigenvectors and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$, the eigenvalue λ_i can be interpreted as the graph frequency corresponding to the basis vector v_i . Graph frequencies are ordered according to the total variation of the eigenvectors. Following Sandryhaila and Moura (2014) and Singh et al. (2016), eigenvectors with smaller total variation are considered low-frequency components, while those with larger variation are regarded as high-frequency components. In particular, when $S = L$, the graph frequencies are real-valued, and their ordering coincides with the usual numerical ordering.

Given the eigendecomposition $S = V\Lambda V^H$ with eigenvalues ordered from low to high frequencies, the graph Fourier transform (GFT) of a signal x is defined as $\tilde{x} = V^H x$. The ℓ th element, $\tilde{x}(\lambda_\ell) = v_\ell^H x$, is the graph Fourier coefficient of x at graph frequency λ_ℓ . The GFT therefore transforms a signal from the node domain to the graph frequency (or spectral) domain, analogous to the classical Fourier transform. A linear and shift-invariant (LSI) graph filter is a linear operator that modifies graph spectral components in a frequency-dependent manner. It can be written as $H = \sum_{\ell=1}^n h_\ell S^{\ell-1} = V \text{diag}(\{h(\lambda_\ell)\}_{\ell=1}^n) V^H$, where $h(\lambda) = \sum_{\ell=1}^n h_\ell \lambda^{\ell-1}$ denotes the frequency response of the filter (Sandryhaila and Moura 2013). The filtered output of x is then Hx . These filters are essential in GSP for tasks such as denoising, smoothing, and signal reconstruction on graphs (Tremblay et al. 2018). In this paper, the term ‘graph filter’ specifically refers to LSI graph filters unless stated otherwise.

2.2 Stationarity and Spectral Densities on Graphs

The concept of weak stationarity for random processes on graphs has been widely used in statistical graph signal processing (Kim and Oh 2025a, Perraudin and Vandergheynst 2017, Marques et al. 2017). The definition of weak stationarity for multivariate random graph processes below was introduced by Kim and Oh (2024) as an extension of the univariate notion in Marques et al. (2017).

Definition 2.1 .

(Kim and Oh 2024) Given a normal graph shift operator S , consider a zero-mean p -dimensional random graph process $X = (X_1 \mid \dots \mid X_p)$, where each X_i ($1 \leq i \leq p$) is an n -dimensional vector. The random graph process X is said to be weakly stationary if the cross-covariance matrices $\Sigma_{ij}^X := \text{Cov}(X_i, X_j)$ and S are simultaneously diagonalizable for $1 \leq i, j \leq p$.

If X_i ($1 \leq i \leq p$) are not zero-mean random graph processes, their first moments are required to satisfy $\mathbb{E}[X_i] \propto v_j$ ($1 \leq j \leq n$). This definition is relative to the chosen GSO S , which serves as a signal shift on a graph. It implies that the covariance and cross-covariance matrices are diagonalized by the graph Fourier basis associated with S . This is analogous to the time-series case, in which the covariance structure of a weakly stationary process is diagonalized by the standard Fourier basis. Moreover, as discussed in Kim and Oh (2025a), Definition 2.1 is equivalent to a graph-domain shift-invariance condition. This equivalence links graph stationarity to the familiar time-series concept of weak stationarity, in which second-order moments are invariant to time shifts and depend only on relative lags.

Although there is currently no widely accepted formal statistical test for graph stationarity, diagnostic measures have been proposed to quantify how closely a graph process satisfies weak stationarity with respect to a given GSO (Perraudin and Vandergheynst 2017, Marques et al. 2017, Kim and Oh 2025a). These diagnostics are based on the degree of diagonal concentration of the estimated covariance or cross-covariance structure in the graph Fourier domain.

For a weakly stationary random graph process X , Kim and Oh (2025a) defined the graph cross-spectral density (GCSD) between X_i and X_j ($1 \leq i, j \leq p$) as an $n \times 1$ vector,

$p_{ij}^X = (p_{ij}^X(\lambda_1), \dots, p_{ij}^X(\lambda_n))^T = \text{diag}(V^H \Sigma_{ij}^X V)$. GCSD characterizes the frequency-wise dependence between two graph signals. When $i = j$, this becomes the graph power spectral density (GPSD) of X_i introduced in Marques et al. (2017), which shows the distribution of signal energy across graph frequencies. Both GPSD and GCSD can be estimated using the windowed average graph periodogram and graph cross-periodogram, respectively, with random windows applicable in practical implementations (Kim and Oh 2025a). The (squared) graph coherence between X_i and X_j ($1 \leq i, j \leq p$) was subsequently defined in Kim and Oh (2025a) as an $n \times 1$ vector $c_{ij}^X = (c_{ij}^X(\lambda_1), \dots, c_{ij}^X(\lambda_n))^T$, where

$$c_{ij}^X(\lambda_\ell) = \frac{|p_{ij}^X(\lambda_\ell)|^2}{p_{ii}^X(\lambda_\ell)p_{jj}^X(\lambda_\ell)}, \quad 1 \leq i, j \leq p, 1 \leq \ell \leq n.$$

For brevity, ‘weak stationarity’ will hereafter be called ‘stationarity,’ and ‘(squared) graph coherence’ as ‘graph coherence.’ The term ‘graph signal’ is used interchangeably to refer to either a random graph process or its realization, depending on the context.

2.3 Related Work

Several studies have integrated graph structures into CCA. Yuan and Sun (2014) proposed graph-regularized multiset canonical correlation analysis (GrMCCA) for multiple datasets measured on the same observations across multiple classes. For each dataset, they construct k -nearest-neighbor within-class and between-class graphs from observation similarities. These graphs are incorporated into the CCA objective to maximize correlations between canonical variables while promoting within-class compactness and between-class separation. Chen et al. (2018) proposed graph-regularized CCA (gCCA), which assumes low-dimensional common

latent sources generating the two datasets, aligning with the perspective of Bach and Jordan (2005). The latent sources are assumed to form a smooth multivariate graph signal $s = (s_1, \dots, s_n)^\top$, where $s_i \in \mathbb{R}^p$, on a source graph $\mathcal{G}_s$, whose nodes represent the n observations. The source graph may be given or constructed from prior knowledge. Since the canonical variables α_i and β_j serve as one-dimensional approximations of s_i and s_j , respectively, the canonical variable vectors $\alpha = (\alpha_1, \dots, \alpha_n)^\top$ and $\beta = (\beta_1, \dots, \beta_n)^\top$ are obtained by maximizing $\text{Corr}(\alpha, \beta) - \gamma \alpha^\top L_s \beta$, where L_s denotes the graph Laplacian of $\mathcal{G}_s$. Chen et al. (2019) further extended gCCA to multiview settings, resulting in MGCCA. These methods operate in the node domain and impose smoothness constraints on the canonical variables. While this assumption is often valid, it does not always apply. In contrast, the proposed method incorporates graph structure in the frequency domain, providing a more general framework beyond cases where canonical variables on neighboring nodes are expected to have similar values. It produces interpretable canonical variables from a multiscale spectral perspective, offering insights unavailable in existing node-domain approaches. We compare the proposed method with gCCA by Chen et al. (2018) via an image classification task in Section E of the supplement, as it is the most relevant baseline. We further note that the recent graph CCA method of Park et al. (2025) fundamentally differs from the previous methods discussed above and from our method because it uses a bipartite graph whose nodes represent the variables of the two datasets, rather than a graph structure based on observations.

3 Proposed Method

3.1 Graph Canonical Coherence Analysis and Its Properties

Let $X = (X_1 \mid \dots \mid X_p)$ and $Y = (Y_1 \mid \dots \mid Y_q)$ denote p -dimensional and q -dimensional graph signals on a graph $\mathcal{G}$, respectively. We construct an r -dimensional graph signal $Z = (Z_1 \mid \dots \mid Z_r)$ for $r \leq \min(p, q)$, where each component is defined as a sum of graph-filtered outputs of signals $X_1, \dots, X_p$,

$$Z_i = H_{i1}X_1 + \dots + H_{ip}X_p =: \mathcal{T}_i^H(X), \quad 1 \leq i \leq r. \quad (1)$$

Here, $H_{ij} = V \text{diag}(\{h_{ij}(\lambda_\ell)\}_{\ell=1}^n) V^H$ is a graph filter, and $\mathcal{T}_i^{(\cdot)}(\cdot)$ denotes a linear operator that sums graph-filtered outputs. Similarly, we construct an r -dimensional graph signal $W = (W_1 \mid \dots \mid W_r)$, whose components are defined as sums of graph-filtered versions of $Y_1, \dots, Y_q$,

$$W_i = F_{i1}Y_1 + \dots + F_{iq}Y_q =: \mathcal{T}_i^F(Y), \quad 1 \leq i \leq r, \quad (2)$$

where $F_{ij} = V \text{diag}(\{f_{ij}(\lambda_\ell)\}_{\ell=1}^n) V^H$ is a graph filter. We define $h_i(\lambda_\ell) = (h_{i1}(\lambda_\ell), \dots, h_{ip}(\lambda_\ell))^H$ and $f_i(\lambda_\ell) = (f_{i1}(\lambda_\ell), \dots, f_{iq}(\lambda_\ell))^H$.

In a manner similar to classical CCA, we aim to sequentially find Z_i and W_i by maximizing the graph coherence between Z_i and W_i at each graph frequency, subject to the constraint that their graph power spectral densities are equal to 1 across all graph frequencies. Additionally, Z_i and W_i are constrained to have zero coherence with Z_j and W_j for $j=1, \dots, i-1$ across all graph frequencies. The resulting Z_i and W_i are described in Theorem 3.1.

Theorem 3.1 .

Let $XY = (X_1 \mid \dots \mid X_p \mid Y_1 \mid \dots \mid Y_q)$ be a $(p+q)$ -dimensional stationary graph signal with respect to the graph shift operator $S = V\Lambda V^H$ on $\mathcal{G}$. Let p_{ij}^X and p_{ij}^Y denote the graph cross-spectral densities of (X_i, X_j) and (Y_i, Y_j) , respectively, and let p_{ij}^{XY} and p_{ij}^{YX} denote the graph cross-spectral densities of (X_i, Y_j) and (Y_i, X_j) , respectively. For each graph frequency λ_ℓ ($1 \leq \ell \leq n$), define the spectral matrices, $P_X(\lambda_\ell) \in \mathbb{C}^{p \times p}$, $P_Y(\lambda_\ell) \in \mathbb{C}^{q \times q}$, $P_{XY}(\lambda_\ell) \in \mathbb{C}^{p \times q}$, $P_{YX}(\lambda_\ell) \in \mathbb{C}^{q \times p}$, whose (i, j) th elements are $p_{ij}^X(\lambda_\ell)$, $p_{ij}^Y(\lambda_\ell)$, $p_{ij}^{XY}(\lambda_\ell)$, and $p_{ij}^{YX}(\lambda_\ell)$, respectively. Suppose that $P_X(\lambda_\ell)$ and $P_Y(\lambda_\ell)$ are nonsingular for all $1 \leq \ell \leq n$.

The solutions maximizing the graph coherence of Z_i and W_i across each graph frequency, subject to the constraints that (i) the graph power spectral densities of Z_i and W_i are equal to 1 across all frequencies, and (ii) Z_i and W_i have zero coherence with Z_j and W_j for $j=1, \dots, i-1$ across all graph frequencies, are given by

$$\begin{aligned} h_i(\lambda_\ell) &\propto P_X(\lambda_\ell)^{-1} P_{XY}(\lambda_\ell) P_Y(\lambda_\ell)^{-1/2} \eta_i(\lambda_\ell), \quad 1 \leq i \leq r, \\ f_i(\lambda_\ell) &\propto P_Y(\lambda_\ell)^{-1/2} \eta_i(\lambda_\ell), \quad 1 \leq i \leq r, \end{aligned}$$

where $h_i(\lambda_\ell)$ and $f_i(\lambda_\ell)$ are normalized such that $h_i(\lambda_\ell)^H P_X(\lambda_\ell) h_i(\lambda_\ell) = 1$ and $f_i(\lambda_\ell)^H P_Y(\lambda_\ell) f_i(\lambda_\ell) = 1$, thereby satisfying constraint (i). Here, $\eta_i(\lambda_\ell)$ is the i th orthonormal eigenvector of $P_Y(\lambda_\ell)^{-1/2} P_{YX}(\lambda_\ell) P_X(\lambda_\ell)^{-1} P_{XY}(\lambda_\ell) P_Y(\lambda_\ell)^{-1/2}$ corresponding to the i th largest eigenvalue $\gamma_i(\lambda_\ell)$. They satisfy

$$\begin{aligned} P_X(\lambda_\ell)^{-1} P_{XY}(\lambda_\ell) P_Y(\lambda_\ell)^{-1} P_{YX}(\lambda_\ell) h_i(\lambda_\ell) &= \gamma_i(\lambda_\ell) h_i(\lambda_\ell), \\ P_Y(\lambda_\ell)^{-1} P_{YX}(\lambda_\ell) P_X(\lambda_\ell)^{-1} P_{XY}(\lambda_\ell) f_i(\lambda_\ell) &= \gamma_i(\lambda_\ell) f_i(\lambda_\ell). \end{aligned}$$

The maximum coherence between Z_i and W_i is $\gamma_i(\lambda_\ell)$. Moreover, Z_i and W_i have zero coherence with Z_j and W_j for $j < i$.

A proof is provided in Section B of the supplement. The pair of Z_i and W_i in Theorem 3.1 is called the i th pair of (population) canonical graph signals. Their graph coherence, $\gamma_i(\lambda_\ell)$, is

called the i th (population) graph canonical coherence. The entire procedure for implementing CCA on graphs is referred to as ‘gCChA’. This procedure can alternatively be formulated from an error minimization perspective, as described in Section A of the supplement.

To implement gCChA, the detailed procedure is outlined in Algorithm C.1 of the supplement, which provides sample versions of the canonical graph signal pairs and graph canonical coherences, denoted by $\{(\hat{Z}_i, \hat{W}_i)\}_{i=1}^r$ and $\hat{\gamma}_i(\lambda_\ell)$. The proposed method is implemented by repeating the procedure for the population canonical graph signals, replacing the population GCSDs with their estimates, since the true GCSDs are unknown. For this, one can use either the graph cross-periodogram or the windowed average graph cross-periodogram proposed by Kim and Oh (2025a). If multiple realizations of the two sets of graph signals X and Y are available, the graph cross-periodogram is used. If only a small number of realizations are available, the windowed average graph cross-periodogram with random windows is used.

3.2 Interpretation of Graph Canonical Coherence Analysis

In classical CCA, interpretation is typically based on correlations between the original variables and their respective canonical variates, known as canonical loadings, rather than on the canonical coefficients themselves, which can be unstable in the presence of multicollinearity. Furthermore, correlations between the original variables and the canonical variates of the opposite set, called canonical cross-loadings, can be used to express the relationship between a variable in one set and a canonical variate in the other. In gCChA, we similarly compute the graph coherences between the original graph signals in each dimension and the respective or opposite canonical graph signals, called *graph canonical loadings* and *graph canonical cross-loadings*, respectively. These quantities facilitate interpretation in the graph frequency domain, representing the proportion of variance in the graph Fourier coefficients of each variable that is linearly shared with the canonical graph signals from either the same set or the opposite set. Because the graph coherences contain information at each graph frequency λ_ℓ , we can provide a multiscale interpretation across graph frequencies. For example, at a low graph frequency, a high graph coherence value between a canonical graph signal and a particular original signal indicates that they are closely related at a coarse scale. Conversely, at high graph frequency, a high graph coherence value indicates a close relationship at fine scales. An example of a more detailed interpretation is presented in Section 4.2.

Motivated by well-established terminology in classical CCA, we introduce similar quantities in the gCChA framework. The *graph canonical communality*, defined as the sum of graph canonical loadings for each canonical graph signal, indicates how much of the variance in each variable’s graph Fourier coefficients can be explained by the canonical graph signals. In other words, it reflects the importance of each variable in establishing the canonical relationship between the two sets within the graph frequency domain. Specifically, the graph canonical

communality of X_j at graph frequency λ_ℓ , is given by $\sum_{i=1}^r c_{ij}^{ZX}(\lambda_\ell)$, and that of Y_j is $\sum_{i=1}^r c_{ij}^{WY}(\lambda_\ell)$,

where $c_{ij}^{ZX}(\lambda_\ell)$ and $c_{ij}^{WY}(\lambda_\ell)$ represent the coherences between Z_i and X_j and between W_i and Y_j , respectively, at graph frequency λ_ℓ , respectively, defined by

$$c_{ij}^{ZX}(\lambda_\ell) = \frac{|p_{ij}^{ZX}(\lambda_\ell)|^2}{p_{ii}^Z(\lambda_\ell)p_{jj}^X(\lambda_\ell)} \quad \text{and} \quad c_{ij}^{WY}(\lambda_\ell) = \frac{|p_{ij}^{WY}(\lambda_\ell)|^2}{p_{ii}^W(\lambda_\ell)p_{jj}^Y(\lambda_\ell)}.$$

To evaluate the explanatory power of the canonical graph signals at each graph frequency λ_ℓ , we define the *graph canonical adequacy* as the average of all graph canonical loadings for the original graph signals in a given set with respect to a single canonical graph signal. This quantity reflects how well the canonical graph signals summarize the variability of each set of graph signals at each frequency in the graph frequency domain. Specifically, the graph canonical

adequacy of Z_i at graph frequency λ_ℓ is given by $\sum_{j=1}^p c_{ij}^{ZX}(\lambda_\ell) / p$, and that of W_i is $\sum_{j=1}^q c_{ij}^{WY}(\lambda_\ell) / q$.

. As shown in Proposition 3.1, the graph canonical communalities range from 0 to 1, and the cumulative explanatory power, obtained by summing the graph canonical adequacies up to each component, also falls between 0 and 1. At a given graph frequency λ_ℓ , the cumulative explanatory power of the first t ($t \leq r$) canonical graph signals $Z_1, \dots, Z_t$, defined as

$\sum_{i=1}^t \sum_{j=1}^p c_{ij}^{ZX}(\lambda_\ell) / p$, represents the proportion of the total variance explained by these canonical graph signals at frequency λ_ℓ in the graph frequency domain. Similarly, the cumulative

explanatory power of the first t canonical graph signals $W_1, \dots, W_t$, given by $\sum_{i=1}^t \sum_{j=1}^q c_{ij}^{WY}(\lambda_\ell) / q$, quantifies the proportion of variance explained by these canonical graph signals at frequency λ_ℓ .

Proposition 3.1 .

For $1 \leq \ell \leq n$, the following inequalities hold:

$$\begin{aligned} 0 &\leq \sum_{i=1}^r c_{ij}^{ZX}(\lambda_\ell) \leq 1, j = 1, \dots, p, \quad \text{and} \quad 0 \leq \sum_{i=1}^r c_{ij}^{WY}(\lambda_\ell) \leq 1, j = 1, \dots, q. \quad \text{Consequently,} \\ 0 &\leq \frac{1}{p} \sum_{i=1}^r \sum_{j=1}^p c_{ij}^{ZX}(\lambda_\ell) \leq 1, \quad \text{and} \quad 0 \leq \frac{1}{q} \sum_{i=1}^r \sum_{j=1}^q c_{ij}^{WY}(\lambda_\ell) \leq 1. \quad \text{Furthermore, if } r = \min(p, q), \text{ then} \\ \sum_{i=1}^r c_{ij}^{ZX}(\lambda_\ell) &= 1 \text{ when } p < q, \quad \text{and} \quad \sum_{i=1}^r c_{ij}^{WY}(\lambda_\ell) = 1 \text{ when } p > q. \end{aligned}$$

A proof is provided in Section B of the supplement. We note that both the interpretations and the proposition in this section directly apply to the sample versions.

4 Numerical Experiments

4.1 Simulation Study

We conduct a simulation study to evaluate whether the proposed gCChA method can recover population graph canonical coherences in a setting where the underlying graph and the target

graph-frequency quantities are fixed and explicitly defined. To this end, we generate a connected random geometric graph with $n=50$ nodes. The nodes are independently sampled from the unit square, and two nodes are connected if their Euclidean distance is below a prespecified radius. The weight of an edge between nodes u and v is $w_{uv} = \exp\{d^2(u,v)/\bar{d}^2\}$, where $d(u,v)$ is the Euclidean distance between the two nodes and $\bar{d}$ is the average pairwise distance. The graph Laplacian serves as the graph shift operator, with eigenvalues $\lambda_1 \leq \dots \leq \lambda_n$. The graph and its associated graph Fourier basis remain fixed throughout the simulation.

We consider two bivariate graph signals, $X = (X_1 \mid X_2)$ and $Y = (Y_1 \mid Y_2)$. Let $P_X(\lambda_\ell)$, $P_Y(\lambda_\ell)$, and $P_{XY}(\lambda_\ell)$ denote the population spectral matrices of X , Y , and their cross-spectral matrix, respectively, at graph frequency λ_ℓ . We examine two scenarios: (i) diagonal within-set spectral matrices, in which $P_X(\lambda_\ell)$ and $P_Y(\lambda_\ell)$ are diagonal, and (ii) correlated within-set spectral matrices, in which $P_X(\lambda_\ell)$ and $P_Y(\lambda_\ell)$ are non-diagonal. In both scenarios, we specify smoothly varying population graph canonical coherences as $\gamma_i(\lambda_\ell) = \rho_i^2(\lambda_\ell)$, $i=1,2$, where $\rho_1(\lambda_\ell) = 0.90 - 0.25\{1 - \exp(-2\tilde{\lambda}_\ell)\}$ and $\rho_2(\lambda_\ell) = 0.25 + 0.20\{1 - \exp(-2\tilde{\lambda}_\ell)\}$, with $\tilde{\lambda}_\ell = (\lambda_\ell - \lambda_1)/(\lambda_n - \lambda_1)$. By construction, $\rho_1(\lambda_\ell) > \rho_2(\lambda_\ell)$ for all graph frequencies. Specifically, in Scenario 1, we set

$$P_X(\lambda_\ell) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad P_Y(\lambda_\ell) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and $P_{XY}(\lambda_\ell) = \text{diag}\{\rho_1(\lambda_\ell), \rho_2(\lambda_\ell)\}$. In Scenario 2, we introduce within-set spectral dependence by setting

$$P_X(\lambda_\ell) = \begin{pmatrix} 1 & \theta_X(\lambda_\ell) \\ \theta_X(\lambda_\ell) & 1 \end{pmatrix}, \quad P_Y(\lambda_\ell) = \begin{pmatrix} 1 & \theta_Y(\lambda_\ell) \\ \theta_Y(\lambda_\ell) & 1 \end{pmatrix},$$

where $\theta_X(\lambda_\ell) = 0.35\exp(-1.5\tilde{\lambda}_\ell)$ and $\theta_Y(\lambda_\ell) = 0.25 + 0.15\{1 - \exp(-2\tilde{\lambda}_\ell)\}$. The cross-spectral matrix is constructed as $P_{XY}(\lambda_\ell) = P_X(\lambda_\ell)^{1/2} U_1 D(\lambda_\ell) U_2^\top P_Y(\lambda_\ell)^{1/2}$, where $D(\lambda_\ell) = \text{diag}\{\rho_1(\lambda_\ell), \rho_2(\lambda_\ell)\}$, and U_1 and U_2 are rotation matrices with rotation angles $\pi/6$ and $\pi/5$, respectively. In both scenarios, by Theorem 3.1, the corresponding population graph canonical coherences satisfy $\gamma_i(\lambda_\ell) = \rho_i^2(\lambda_\ell)$.

Under weak stationarity, the variances and covariances of the graph Fourier coefficients at frequency λ_ℓ are given by the corresponding graph power and cross-spectral densities defined in Section 2.2. Accordingly, the joint covariance structure of the graph Fourier coefficients can be specified by $P_X(\lambda_\ell)$, $P_Y(\lambda_\ell)$, and $P_{XY}(\lambda_\ell)$. For each graph frequency λ_ℓ , we generate R independent realizations in the graph frequency domain according to

$$\begin{pmatrix} \tilde{X}^{(r)}(\lambda_\ell) \\ \tilde{Y}^{(r)}(\lambda_\ell) \end{pmatrix} \sim N \left(0, \begin{pmatrix} P_X(\lambda_\ell) & P_{XY}(\lambda_\ell) \\ P_{YX}(\lambda_\ell) & P_Y(\lambda_\ell) \end{pmatrix} \right), \quad r=1, \dots, R,$$

where $\tilde{X}^{(r)}(\lambda_\ell) = (\tilde{X}_1^{(r)}(\lambda_\ell), \tilde{X}_2^{(r)}(\lambda_\ell))^\top$ and $\tilde{Y}^{(r)}(\lambda_\ell) = (\tilde{Y}_1^{(r)}(\lambda_\ell), \tilde{Y}_2^{(r)}(\lambda_\ell))^\top$, whose entries represent the graph Fourier coefficients of the corresponding components at graph frequency λ_ℓ . The graph Fourier coefficient vectors are generated independently for each distinct graph frequency and then transformed to the node domain using the inverse graph Fourier transform, yielding $X^{(r)} = (X_1^{(r)} \mid X_2^{(r)})$ and $Y^{(r)} = (Y_1^{(r)} \mid Y_2^{(r)})$, $r=1, \dots, R$. By construction, the resulting graph signals satisfy the weak stationarity condition in Definition 2.1. For each $R \in \{50, 100, 200, 400\}$, the simulation is repeated 100 times. In each repetition, the estimation accuracy of the i th graph canonical coherence is evaluated using the mean squared error (MSE), defined as $\sum_{\ell=1}^n \{\hat{\gamma}_i(\lambda_\ell) - \gamma_i(\lambda_\ell)\}^2 / n$, where $\hat{\gamma}_i(\lambda_\ell)$ denotes the estimated graph canonical coherence.

Figure 2 compares the estimated graph canonical coherences with their population counterparts for a representative simulation run under the two scenarios. The curves are based on $R = 200$ realizations. In both scenarios, the estimated curves closely follow the overall frequency-dependent patterns of the population graph canonical coherences. The first graph canonical coherence decreases from low to high graph frequencies, whereas the second increases. These contrasting patterns are successfully recovered by the proposed method. Figure 3 reports the average MSEs across 100 replications, with error bars indicating one standard deviation. Across both simulation settings and both canonical components, the MSE decreases steadily as the number of realizations R increases. The results are similar across the two scenarios, indicating that estimation accuracy is not substantially affected by within-set spectral correlations. The standard deviations likewise decrease as R increases. Overall, these results demonstrate that gCChA accurately recovers the population graph canonical coherences on a fixed underlying graph and that estimation accuracy improves as more realizations become available.

4.2 G20 Network Data Analysis

We apply gCChA to data from the G20 network. We construct a graph with 18 nodes representing G20 countries, excluding Argentina and the European Union due to missing data. The edge weights are defined as $\log(1 + \text{trade asset})$ based on trade data from 2019 (Kim and Oh 2025b). The resulting trading network is shown in Figure 4, and the country codes are listed in Table D.1 of the supplement.

Two sets of observed graph signals, X and Y , denote economic and energy-related indicators, respectively. Each set consists of five graph signals, denoted as $X = (X_1 \mid \dots \mid X_5)$ and $Y = (Y_1 \mid \dots \mid Y_5)$, which correspond to (exports, foreign direct investment, GDP growth, imports, inflation) and (energy imports, energy use, fossil fuel consumption, GDP per unit of energy use, renewable energy consumption). Brief descriptions of the graph signals are provided in Table D.2 of the supplement, with detailed definitions available in the World Bank Group's metadata

glossary. The data are from the World Bank Open Data website (<https://data.worldbank.org/indicator>). In summary, we have $n=18$, $p=5$, and $q=5$.

In implementing gCChA, we use the graph Laplacian L as GSO. The estimation of the graph power and cross-spectral densities is performed using the windowed averaging method with 50 random windows (Kim and Oh 2025a). There are five pairs of canonical graph signals (i.e., $r = \min(p, q) = 5$), $(\hat{Z}_1, \hat{W}_1), \dots, (\hat{Z}_5, \hat{W}_5)$. The graph canonical coherences within each pair, $\hat{\gamma}_1(\lambda_\ell), \dots, \hat{\gamma}_5(\lambda_\ell)$, are shown in Figure 5, where λ_ℓ denotes the graph frequencies of the trading network. As expected, the canonical coherences decrease from $(\hat{Z}_1, \hat{W}_1)$ to $(\hat{Z}_5, \hat{W}_5)$ across all graph frequencies.

From Figure 5, we observe a noticeable gap between the graph canonical coherences of $(\hat{Z}_2, \hat{W}_2)$ and $(\hat{Z}_3, \hat{W}_3)$. We calculate the cumulative explanatory powers of the first two canonical graph signals, $\sum_{i=1}^2 \sum_{j=1}^p c_{ij}^{\hat{Z}\hat{X}}(\lambda_\ell) / p$ and $\sum_{i=1}^2 \sum_{j=1}^q c_{ij}^{\hat{W}\hat{Y}}(\lambda_\ell) / q$, which exceed 0.75, indicating that more than 75% of the variance within X and Y , on average across graph frequencies, is explained by the first two respective canonical graph signals. We therefore interpret the first two canonical pairs. Although interpretation can be applied at each graph frequency, we examine λ_1 and λ_7 as representative frequencies for coarse and fine scales, respectively. Figure 6 shows the associated basis signals v_1 and v_7 . The basis signal v_1 is constant across all nodes, representing the coarsest-scale pattern of the G20 network. In contrast, v_7 highlights the contrast between Germany and the United States.

At the lowest frequency λ_1 , the canonical graph signals depict broad relationships across the G20 network, capturing country-level similarity patterns that dominate at a global scale, as reflected by the constant basis signal v_1 . At this frequency, as shown in the top row of Figure 7, $\hat{Z}_1$ is associated with X_1 , X_3 , and X_4 in the $(-), (+), (-)$ directions, while $\hat{W}_1$ is associated with Y_1 and Y_5 in the $(-)$ and $(+)$ directions, respectively. Therefore, the first canonical pair $(\hat{Z}_1, \hat{W}_1)$ shows that countries with higher GDP growth but lower trade volumes tend to have lower energy import dependence and greater renewable energy use. Countries such as France, the United Kingdom, Canada, Australia, and Brazil exhibit this pattern, reflecting growth- and sustainability-focused systems characterized by moderate trade activity, steady growth, and a relatively high level of energy independence supported by renewables, which tends to increase both $\hat{Z}_1$ and $\hat{W}_1$. In contrast, export- and industry-driven countries like Korea, Japan, Italy, India, and Türkiye show the opposite trend, causing both $\hat{Z}_1$ and $\hat{W}_1$ to decrease. This reflects greater trade dependence and higher reliance on energy imports, and lower adoption of renewable energy. This canonical relationship captures a broad, global-scale trade-energy connection across the G20 network, as the correlation between $\hat{Z}_1$ and $\hat{W}_1$ at frequency λ_1 reflects a consistent structural pattern shared by many countries. For the second canonical pair, $\hat{Z}_2$ is associated with

X_2 and X_5 , and $\hat{W}_2$ with Y_1 and Y_5 , all in the (+) direction. This pair indicates that countries with higher levels of foreign direct investment (FDI) and inflation tend to have greater energy imports and increased use of renewable energy. This reflects a pattern in which economies with greater openness and inflationary pressures simultaneously increase energy imports and pursue energy transition policies. Countries such as China and India, known for rapid industrial growth, substantial FDI inflows, and ongoing investments in renewable energy, tend to push both $\hat{Z}_2$ and $\hat{W}_2$ higher. In contrast, Canada, Australia, and Saudi Arabia, which are resource-based and domestically stable economies with low inflation and limited reliance on imported energy, tend to pull these variables lower. Overall, the coarse-scale canonical signals reveal the main relationships among economic openness, price trends, and the energy transition observed across the G20 network.

At the high frequency λ_7 , the canonical graph signals highlight fine-scale country contrasts within the G20 trade–energy network, particularly between Germany and the United States, as illustrated by the corresponding basis signal v_7 . For the first canonical pair, $\hat{Z}_1$ is associated with X_2 and X_5 in the (–) and (+) directions, while $\hat{W}_1$ is associated with Y_2 in the (–) direction. This relationship indicates that countries with lower FDI inflows and total energy consumption tend to have higher values of $(\hat{Z}_1, \hat{W}_1)$. Germany, known for relatively modest FDI inflows but efficient and moderate energy use, tends to have higher values of $\hat{Z}_1$ and $\hat{W}_1$. In contrast, the United States, with higher FDI inflows and energy consumption, tends to have smaller values of $\hat{Z}_1$ and $\hat{W}_1$. The second canonical pair exhibits a more complex structure: $\hat{Z}_2$ is associated with X_1, X_2, X_3, X_4 , and X_5 in the (+), (–), (–), (+), (+) directions, while $\hat{W}_2$ is connected to Y_1, Y_2, Y_4 , and Y_5 in the (+), (–), (+), (+) directions, respectively. Unlike the first pair, which mainly shows the connection between capital inflows and energy use, this pair illustrates interconnected relationships among trade dependence, economic growth, and renewable energy strategies. Germany, known for its high trade openness, moderate GDP growth, reliance on imported energy, and strong energy efficiency, tends to have larger values of $\hat{Z}_2$ and $\hat{W}_2$. In contrast, the United States, often described as having lower trade dependence, higher GDP growth, and less reliance on energy imports, tends to have lower values of $\hat{Z}_2$ and $\hat{W}_2$. Overall, the high-frequency canonical signals at λ_7 emphasize localized variations distinguishing advanced economies with differing structural balances among industrial openness, energy efficiency, and policy direction.

5 Concluding Remarks

In this paper, we propose a graph canonical coherence analysis (gCChA) method that operates in the graph frequency domain. The proposed method handles two multivariate graph signals defined on the same graph and produces two canonical graph signals that serve as low-dimensional representations summarizing the dependence structure between the two and capturing their shared information. The proposed framework offers a spectral interpretation of

canonical graph signals by associating them with the original graph signals at each graph frequency.

There are several directions for future work. One avenue is to extend the current framework to accommodate more than two graph datasets, enabling canonical dependence analysis among multiple graph signals. Another direction is to explore different applications with multiple graph-structured datasets, such as brain functional networks and environmental sensing networks. Further methodological development is also of interest, including improved GCSD estimation and optimal window design for the windowed-average graph cross-periodogram. Moreover, although spectral-domain analysis is central to interpreting the proposed gCChA framework, it requires diagonalizing the GSO, which can be computationally demanding for large-scale graphs. Fast and approximate graph Fourier transform techniques (Le Magoarou and Gribonval 2016, Le Magoarou et al. 2017, Lu and Ortega 2019) may provide useful directions for reducing the computational burden in large-scale applications.

A further issue concerns the sensitivity of graph spectral quantities to the choice of GSO. In graph signal processing, the graph and GSO are typically treated as part of the pre-specified data domain, so spectral analysis is interpreted relative to this choice. Accordingly, gCChA framework should be understood conditionally on the specified graph structure and GSO. Since different graph constructions or edge-weight perturbations may affect the graph Fourier basis, the resulting graph spectral density matrices and canonical coherences may also change. Developing perturbation analyses and sensitivity assessments for alternative GSO choices would be valuable future work, with studies on graph signal processing under graph perturbations (Ceci and Barbarossa 2020) serving as a useful starting point. Finally, relaxing the weak stationarity assumption is an important extension. In particular, locally stationary graph processes (Canbolat and Vural 2024) may provide a useful foundation for modeling graph signals with node-varying spectral quantities. Developing a multivariate locally stationary graph process framework, along with estimation procedures for node-varying spectral quantities, would broaden the applicability of gCChA to heterogeneous graph-structured data where the stationarity assumption may be questionable. We provide a brief sketch of a multivariate locally stationary extension in Section F of the supplement. We leave these topics for future investigation.

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Data Availability Statement

The data in this paper are available at <https://github.com/qsoon/gCChA>.

References

- Bach, F. R. and Jordan, M. I. (2005), A probabilistic interpretation of canonical correlation analysis, Technical Report 688, Department of Statistics, University of California, Berkeley.
- Brillinger, D. R. (2001), *Time Series: Data Analysis and Theory*, SIAM.
- Canbolat, A. and Vural, E. (2024), ‘Locally stationary graph processes’, *IEEE Transactions on Signal Processing* **72**, 2323–2332.
- Ceci, E. and Barbarossa, S. (2020), ‘Graph signal processing in the presence of topology uncertainties’, *IEEE Transactions on signal processing* **68**, 1558–1573.
- Chen, J., Wang, G. and Giannakis, G. B. (2019), ‘Graph multiview canonical correlation analysis’, *IEEE Transactions on Signal Processing* **67**(11), 2826–2838.
- Chen, J., Wang, G., Shen, Y. and Giannakis, G. B. (2018), ‘Canonical correlation analysis of datasets with a common source graph’, *IEEE Transactions on Signal Processing* **66**(16), 4398–4408.
- Hotelling, H. (1935), ‘The most predictable criterion’, *Journal of Educational Psychology* **26**(2), 139.
- Hotelling, H. (1936), ‘Relations between two sets of variates’, *Biometrika* **28**(3/4), 321–377.
- Kim, K. and Oh, H.-S. (2024), ‘Principal component analysis in the graph frequency domain’, *arXiv preprint arXiv:2410.08422*.
- Kim, K. and Oh, H.-S. (2025a), ‘Cross-spectral analysis of bivariate graph signals’, *IEEE Transactions on Pattern Analysis and Machine Intelligence* **47**(12), 11141–11151.
- Kim, K. and Oh, H.-S. (2025b), ‘Graph frequency-domain factor modeling’, *IEEE Transactions on Pattern Analysis and Machine Intelligence* pp. 1–13.
- Le Magoarou, L. and Gribonval, R. (2016), ‘Flexible multilayer sparse approximations of matrices and applications’, *IEEE Journal of Selected Topics in Signal Processing* **10**(4), 688–700.
- Le Magoarou, L., Gribonval, R. and Tremblay, N. (2017), ‘Approximate fast graph fourier transforms via multilayer sparse approximations’, *IEEE Transactions on Signal and Information Processing over Networks* **4**(2), 407–420.
- Leus, G., Marques, A. G., Moura, J. M., Ortega, A. and Shuman, D. I. (2023), ‘Graph signal processing: History, development, impact, and outlook’, *IEEE Signal Processing Magazine* **40**(4), 49–60.

Lu, K.-S. and Ortega, A. (2019), ‘Fast graph fourier transforms based on graph symmetry and bipartition’, *IEEE Transactions on Signal Processing* **67**(18), 4855–4869.

Marques, A. G., Segarra, S., Leus, G. and Ribeiro, A. (2017), ‘Stationary graph processes and spectral estimation’, *IEEE Transactions on Signal Processing* **65**(22), 5911–5926.

Park, H., Bai, S., Ye, Z., Lee, H., Ma, T. and Chen, S. (2025), ‘Graph canonical correlation analysis’, *arXiv preprint arXiv:2502.01780* .

Perraudin, N. and Vandergheynst, P. (2017), ‘Stationary signal processing on graphs’, *IEEE Transactions on Signal Processing* **65**(13), 3462–3477.

Sandryhaila, A. and Moura, J. M. (2013), ‘Discrete signal processing on graphs’, *IEEE Transactions on Signal Processing* **61**(7), 1644–1656.

Sandryhaila, A. and Moura, J. M. (2014), ‘Discrete signal processing on graphs: Frequency analysis’, *IEEE Transactions on Signal Processing* **62**(12), 3042–3054.

Segarra, S., Chepuri, S. P., Marques, A. G. and Leus, G. (2018), Statistical graph signal processing: Stationarity and spectral estimation, in P. M. Djurić and C. Richard, eds, ‘Cooperative and Graph Signal Processing’, Elsevier, pp. 325–347.

Shuman, D. I., Narang, S. K., Frossard, P., Ortega, A. and Vandergheynst, P. (2013), ‘The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains’, *IEEE Signal Processing Magazine* **30**(3), 83–98.

Singh, R., Chakraborty, A. and Manoj, B. (2016), Graph fourier transform based on directed Laplacian, in ‘2016 International Conference on Signal Processing and Communications (SPCOM)’, IEEE, pp. 1–5.

Tremblay, N., Gonçalves, P. and Borgnat, P. (2018), Design of graph filters and filterbanks, in ‘Cooperative and Graph Signal Processing’, Elsevier, pp. 299–324.

Yuan, Y.-H. and Sun, Q.-S. (2014), ‘Graph regularized multiset canonical correlations with applications to joint feature extraction’, *Pattern Recognition* **47**(12), 3907–3919.

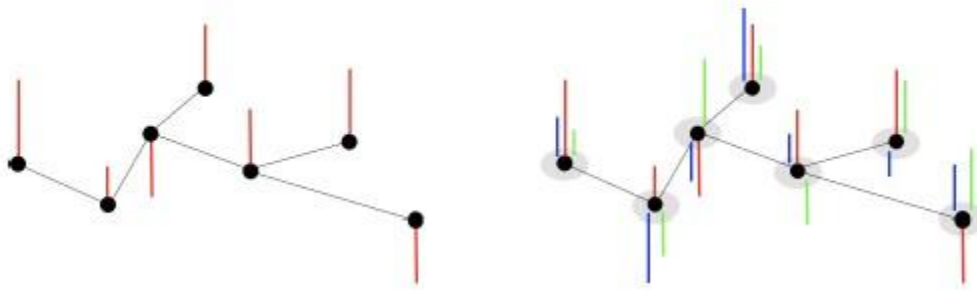


Figure 1: Illustration of univariate (left) and multivariate (right) graph signals.

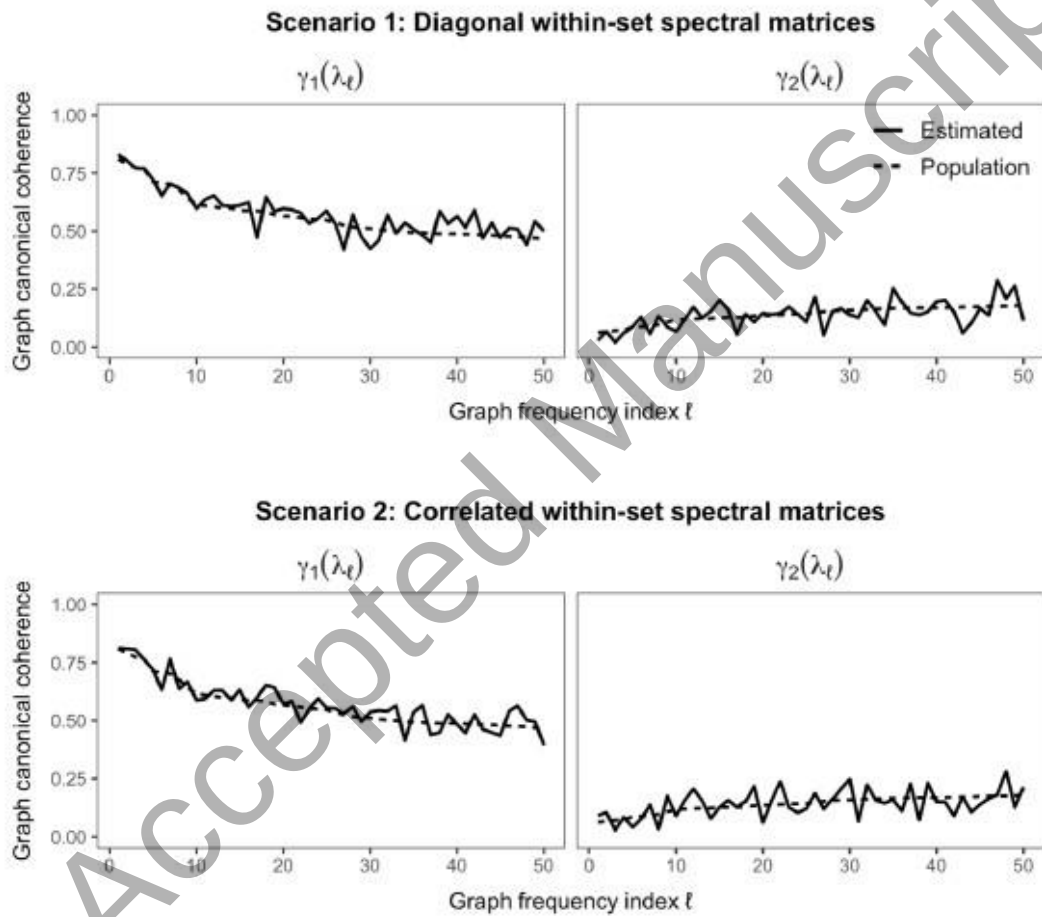


Figure 2: Estimated graph canonical coherence curves and their population counterparts under the two simulation scenarios.

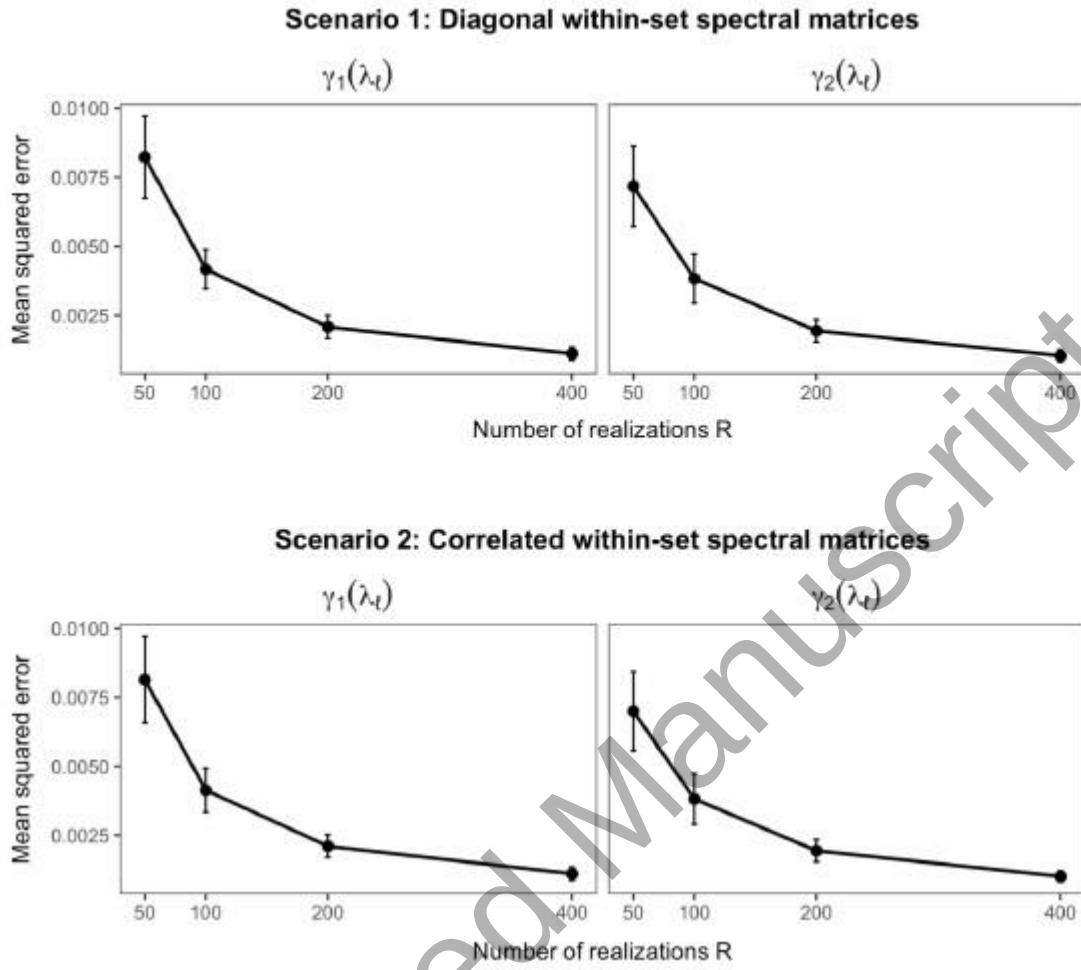


Figure 3: Mean squared error of the estimated graph canonical coherences under the two simulation scenarios. The points and error bars represent the averages and standard deviations over 100 Monte Carlo replications, respectively.

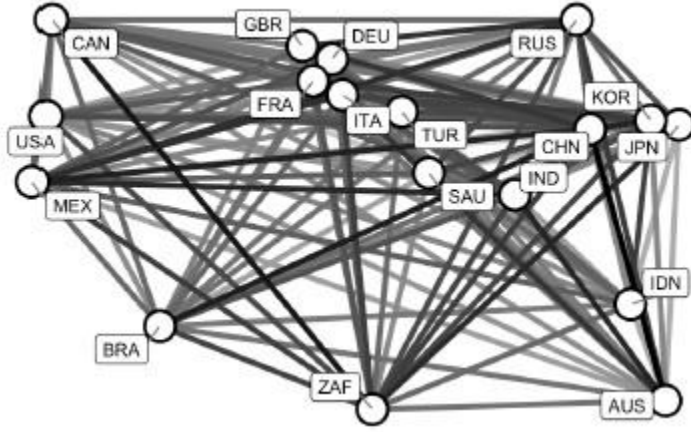


Figure 4: Trading network among G20 countries in 2019, where darker edges represent stronger trade connections.

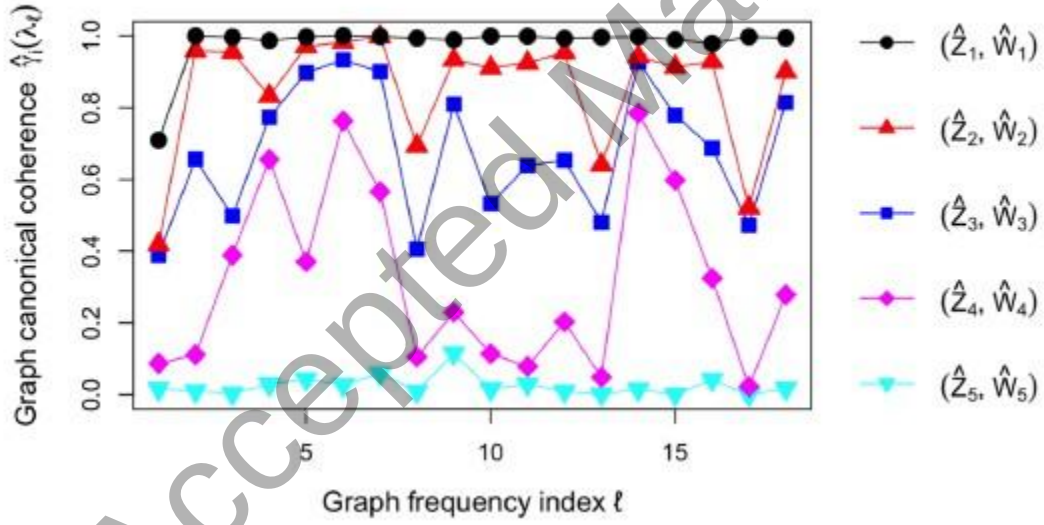


Figure 5: Graph canonical coherences $\hat{\gamma}_i(\lambda_\ell)$ for the canonical graph signal pairs $(\hat{Z}_i, \hat{W}_i)$, $i = 1, \dots, 5$, across graph frequencies.

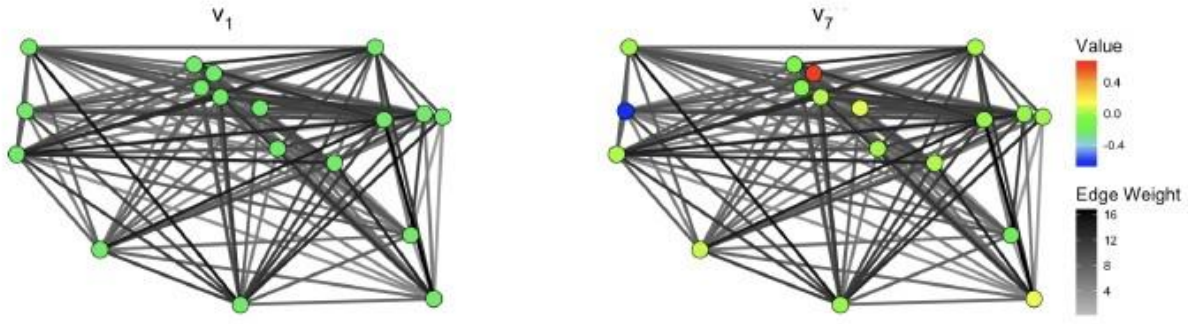


Figure 6: Basis signals v_1 and v_7 .

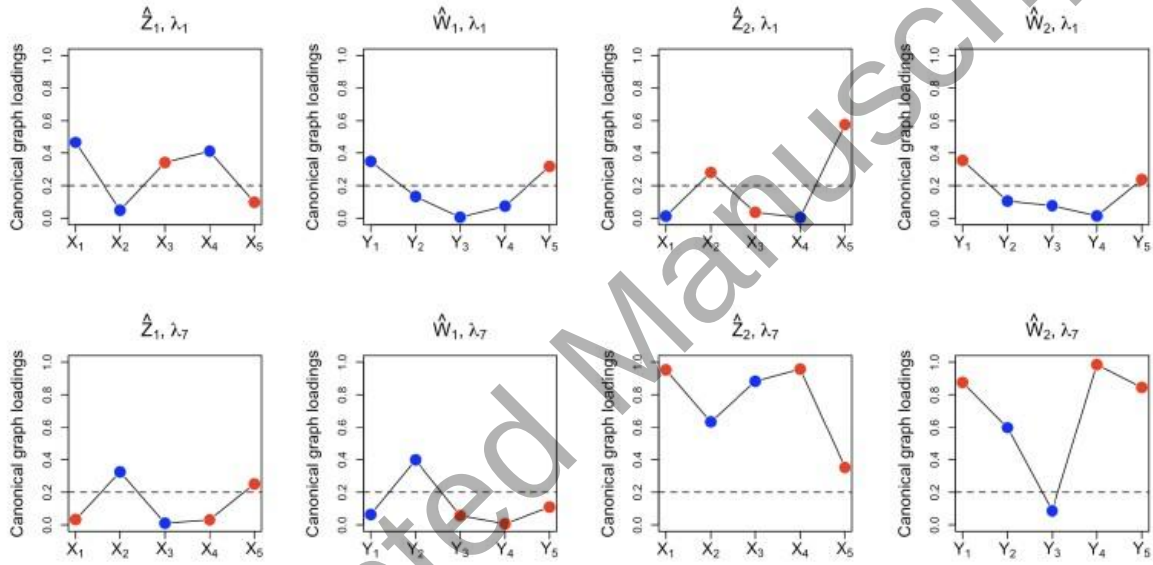


Figure 7: Graph canonical loadings of the first two canonical graph signals at graph frequencies λ_1 (top row) and λ_7 (bottom row). Red dots show positive correlations in the frequency domain, while blue dots represent negative correlations. Loadings with a magnitude exceeding 0.2 (shown by the dashed line) are used for interpretation.